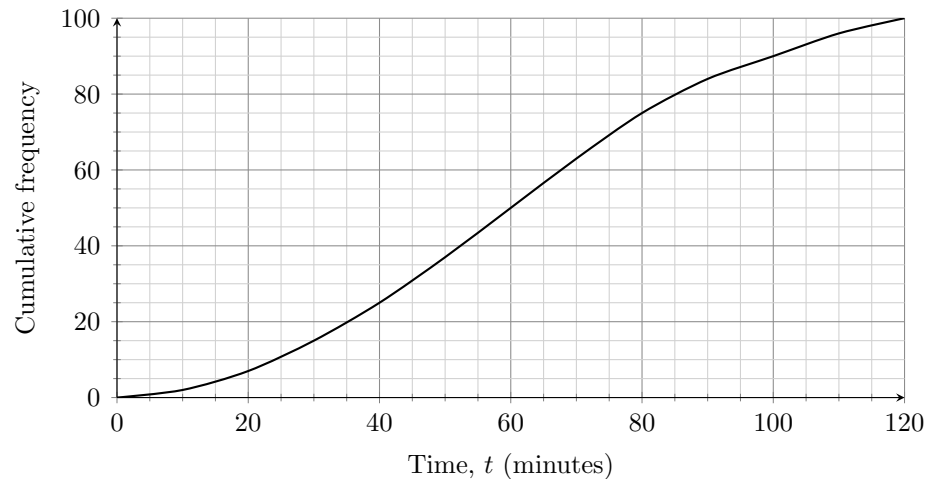


1.7 Classwork: Cumulative frequency graphs

Use your GDC. Draw lines on each graph to show how you read your estimates. An outlier is a value more than $1.5 \times IQR$ below Q_1 or more than $1.5 \times IQR$ above Q_3 .

- The cumulative frequency graph shows the time, t minutes, that 100 students spent on homework last night.



- Write down the median time.
- Find the interquartile range.
- Find the number of students who spent at least 30 minutes on homework.
- The 10% of students who spent the most time on homework are invited to a study-skills workshop. Find the least time that a student invited to the workshop spent on homework.
- The shortest time was 5 minutes and the longest was 118 minutes. Show that there are no outliers in the data.

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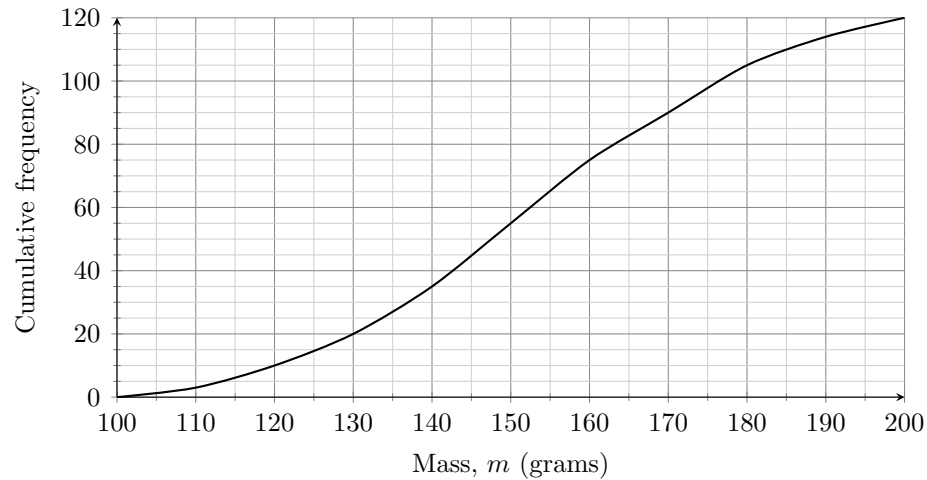
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Nickname:

2. The masses, m grams, of 120 apples are shown in the cumulative frequency graph and summarized in the grouped frequency table.



m (g)	$100 < m \leq 120$	$120 < m \leq 140$	$140 < m \leq 160$	$160 < m \leq 180$	$180 < m \leq 200$
Frequency	10	25	p	q	15

- Estimate the median mass.
- Apples with a mass of more than 170 grams are graded “large”. Find the number of large apples.
- Use the graph to find the value of p and the value of q .
- Find an estimate of the mean mass of the apples.
- Explain why your answer to part (d) is an estimate.